Linear Regression & Logistic Regression

Interview-Style Active Recall Revision

Core concepts, formulas, intuition, assumptions, loss functions, decision boundaries, odds, regularization, and interview questions.

1. Linear Regression

Q1. What is linear regression?

Answer: Linear regression is a supervised learning algorithm used primarily for predicting a continuous numerical target. It models the target as a linear combination of the input features.

Simple: $y\blacksquare = \beta_0 + \beta_1 x$

Multiple: $y\blacksquare = \beta_0 + \beta_1 x_1 + ... + \beta_n x_n$

Q2. What is a residual?

Answer: A residual is the difference between the actual target value and the model's predicted value.

Residual: $e_i = y_i - y\blacksquare_i$

Example: actual = 75, predicted = 70 → residual = +5.
Actual = 65, predicted = 70 → residual = –5.

Q3. What loss function is commonly used for linear regression?

Answer: Mean Squared Error (MSE). It measures the average squared difference between actual and predicted values.

$MSE = (1/n)\ \Sigma_i(y_i - y\blacksquare_i)^2$

Q4. What does ordinary least squares (OLS) do?

Answer: OLS chooses the model coefficients that minimize the sum of squared residuals between the actual and predicted target values.

Minimize: $\Sigma_i(y_i - y\blacksquare_i)^2$

Q5. What are the main assumptions of linear regression?

Answer: 1. Linearity — the relationship between predictors and the expected target is approximately linear.
2. Independence — observations/errors should not be systematically dependent on one another.
3. Homoscedasticity — the variance/spread of residuals is roughly constant across the predictor range.
4. Normality of residuals — residuals are approximately normally distributed, especially important for classical inference, confidence intervals, and hypothesis testing.
5. No severe multicollinearity — predictors should not be excessively correlated with each other.

Q6. What is multicollinearity and why is it a problem?

Answer: Multicollinearity occurs when predictor variables are strongly related. If x2 is approximately a multiple of x1, the model has difficulty isolating the individual effect of each feature. This can make coefficients unstable and difficult to interpret.

Q7. What is R-squared?

Answer: R-squared measures the proportion of variation in the target that is explained by the model relative to simply predicting the mean.

$R^2 = 1 - SS_{res} / SS_{tot}$

$SS_{tot} = \Sigma(y_i - y■)^2$

$SS_{res} = \Sigma(y_i - y■_i)^2$

$SS_{tot} = SS_{exp} + SS_{res}$

Example: $R^2 = 0.90$ means the model explains about 90% of the variation in the target relative to the mean baseline. It does not imply causality or zero prediction error.

Q8. What is adjusted R-squared?

Answer: Adjusted R-squared modifies R-squared by accounting for the number of predictors in the model. Unlike ordinary R-squared, it does not automatically increase simply because additional features are added.

2. Logistic Regression

Q1. What is logistic regression?

Answer: Logistic regression is a supervised learning algorithm used primarily for binary classification. It takes a linear combination of the features and passes the result through a sigmoid function to produce a probability between 0 and 1.

$$z = w^T x + b$$

$$P(Y=1|X) = \sigma(z)$$

Mental model: Logistic regression = linear model + sigmoid.

Q2. What is the sigmoid function and why do we use it?

Answer: The sigmoid function takes any real-valued input and maps it into the range (0, 1). This makes the output useful as a probability estimate for binary classification.

$$\sigma(z) = 1 / (1 + e^{-z})$$

Large positive z → probability approaches 1.
Large negative z → probability approaches 0.
z = 0 → probability = 0.5.

Q3. How does logistic regression make a final class prediction?

Answer: The sigmoid produces a probability, and a threshold is then applied. With the common 0.5 threshold, probabilities at least 0.5 are classified as the positive class and probabilities below 0.5 as the negative class.

Q4. Why is the decision boundary linear even though sigmoid is nonlinear?

Answer: Setting the predicted probability to a threshold alpha gives: sigmoid(w^T x + b) = alpha. Taking the inverse sigmoid gives w^T x + b = log(alpha / (1-alpha)). The right side is a constant, so the boundary is a linear surface.

$$w^T x + b = \log(\alpha / (1-\alpha))$$

For alpha = 0.5, the boundary simplifies to w^T x + b = 0.

Q5. What loss function is used in logistic regression?

Answer: Binary cross-entropy (BCE), also called log loss or negative log-likelihood for the Bernoulli model.

$$BCE = -(1/n)\ \Sigma\ [y_i \log(p_i) + (1-y_i) \log(1-p_i)]$$

Q6. Why does BCE naturally arise from maximum likelihood?

Answer: Logistic regression models each binary target using a Bernoulli distribution. Maximum likelihood tries to maximize the probability of the observed labels. Taking the logarithm converts a product of probabilities into a sum, which is numerically and computationally convenient. Taking the negative gives a quantity we can minimize. This produces binary cross-entropy.

Q7. Why does BCE heavily penalize confidently incorrect predictions?

Answer: For a true positive example, the loss is -log(p). If the model gives a very small probability to the true class, the loss becomes very large. Therefore confidently wrong predictions are strongly penalized.

Q8. What are log-odds and logits?

Answer: The log-odds, also called the logit, are the linear score before the sigmoid. They represent the logarithm of the odds of the positive class.

$$\log(p / (1-p)) = w^T x + b$$

Q9. What are odds?

Answer: Odds compare the probability of the positive class with the probability of the negative class.

$$\text{Odds} = p / (1-p)$$

$$\text{Odds} = e^{w^T x+b}$$

Example: if p = 0.8, odds = 0.8 / 0.2 = 4, meaning 4:1 odds in favor of the positive class.

Q10. How do you interpret a logistic regression coefficient?

Answer: Increasing feature x_j by one unit increases the log-odds by w_j, holding the other features constant. The quantity exp(w_j) gives the multiplicative change in the odds for a one-unit increase in x_j.

$$\Delta \text{ log-odds} = w_j$$

$$O_{new} / O_{old} = e^{w_j}$$

Example: w_j = 0.7 → log-odds increase by 0.7 and odds are multiplied by e^0.7 ≈ 2.01.

Q11. What happens when the classification threshold increases?

Answer: Increasing the threshold makes the classifier more conservative about predicting the positive class. This generally reduces false positives, increasing precision, while increasing false negatives, reducing recall. The number of true positives can also decrease.

Q12. What is regularization and what is the difference between L1 and L2?

Answer: Regularization adds a penalty to the objective to discourage excessively large model coefficients and reduce overfitting.
L1 uses the absolute value of weights and can drive some weights exactly to zero, making it useful for feature selection.
L2 uses squared weights and generally shrinks weights toward zero without making them exactly zero.

$$\text{L1 penalty: } \lambda \Sigma |w_j|$$

$$\text{L2 penalty: } \lambda \Sigma w_j^2$$

3. Quick Interview Cheat Sheet

Linear Regression

• Task: Predict continuous numerical values.
• Model: y_hat = w^T x + b
• Common loss: MSE
• Training: OLS or gradient-based optimization
• Key assumptions: linearity, independence, homoscedasticity, normal residuals for inference, limited multicollinearity.
• R^2: proportion of target variation explained relative to the mean baseline.

Logistic Regression

• Task: Classification, primarily binary classification.
• Linear score: z = w^T x + b
• Probability: sigmoid(z)
• Loss: Binary cross-entropy / negative log-likelihood.
• Decision: apply a threshold to probability.
• Log-odds: log(p/(1-p)) = w^T x + b
• Odds: p/(1-p) = exp(w^T x + b)
• L1: absolute-value penalty; can produce exact zero weights.
• L2: squared penalty; shrinks weights toward zero.

Core mental model:

Linear regression → linear output → continuous prediction.
Logistic regression → linear score → sigmoid → probability → threshold → class.